

Roll no.

MID SEMESTER EXAMINATION SEP 2025

Name of the Program: B. Tech
Name of the Course: Basic Electronics Engineering

Semester: I
Course Code: TEC 101

Time: 90mins

Maximum Marks: 50

Note:

- Answer all the questions by choosing any one of the sub questions.
- Each question carries 10 marks

Q1	(10 marks)	
a)	Convert the following function to its canonical form and find the min-terms and max-terms: $F(A,B,C,D) = (B'.C') + (A'.B.D') + (A.B'.C'.D)$	CO1
b)	OR (i) $(C8D83.0C)_{16} = (?)_{10}$ (ii) $\sqrt{(34)_x} = (5)_x$, find the base "x". (iii) $(3655)_7 = (?)_{10}$ (iv) $(656753)_8 = (?)_{16}$	
Q2	(10 marks)	
a)	Minimize the following function by K map and realize the minimized function by NAND gate only as well as NOR gate only: $F(A, B, C, D) = \sum m(2,3,4,6,10,12,14)$	
	OR	
b)	Define the output 'Q' of the logic circuit given below and write the truth table for the same. Also, find the min-terms and max-terms.	CO1
Q3	(10 marks)	
a)	Calculate the density of impurity atom that must be added to an intrinsic Si crystal to convert it to a: (i) $10^{-4} \Omega\text{m}$ N-type silicon (ii) $10^{-4} \Omega\text{m}$ P-type silicon $\mu_n = 0.138 \text{ m}^2/\text{V-sec}$, and $\mu_p = 0.046 \text{ m}^2/\text{V-sec}$	CO2
	OR	
b)	What is energy band diagram? Explain Classification of materials on the basis of energy band diagram.	
Q4	(10 marks)	CO2

a)	If N_D and N_A are donor and acceptor impurities and n_i is the intrinsic concentration, establish a relation for minority and majority charge densities in extrinsic semiconductors.	
OR		
b)	A sample of Ge has intrinsic charge carrier density of $2.0 \times 10^{19}/m^3$. If it is doped with donor impurity of 1 atom in every 10^7 Ge atom. If the Ge atom density is $4.0 \times 10^{28}/m^3$, then determine the majority and minority charge carriers as well as conductivity and resistivity of the doped semiconductor. Given that the mobility of free electrons and holes is $0.35m^2/V\text{-sec}$ and $0.16m^2/V\text{-sec}$ respectively.	
Q5	(10 marks)	
a)	Simplify following expressions with the help of Boolean algebra: (i) $F=(A+B)(A+B')(A'+C)$ (ii) $F=[(AB)'(C+BD)+A'B']C$ (iii) $F=(C+D)'+A'CD'+AB'C+ACD'$ (iv) $F=(A'B+AB')'(A+B)$	CO1 /CO 2
OR		
b)	A bar of pure Si has a cross sectional area 1.5 mm^2. If the intrinsic concentration is $1.5 \times 10^{16}/m^3$. (i) Determine conductivity if electrons and holes mobility are $0.14 m^2/V\text{-sec}$ and $0.06 m^2/V\text{-sec}$ respectively. (ii) Find the length of the bar if its resistance is $40 \text{ k}\Omega$.	